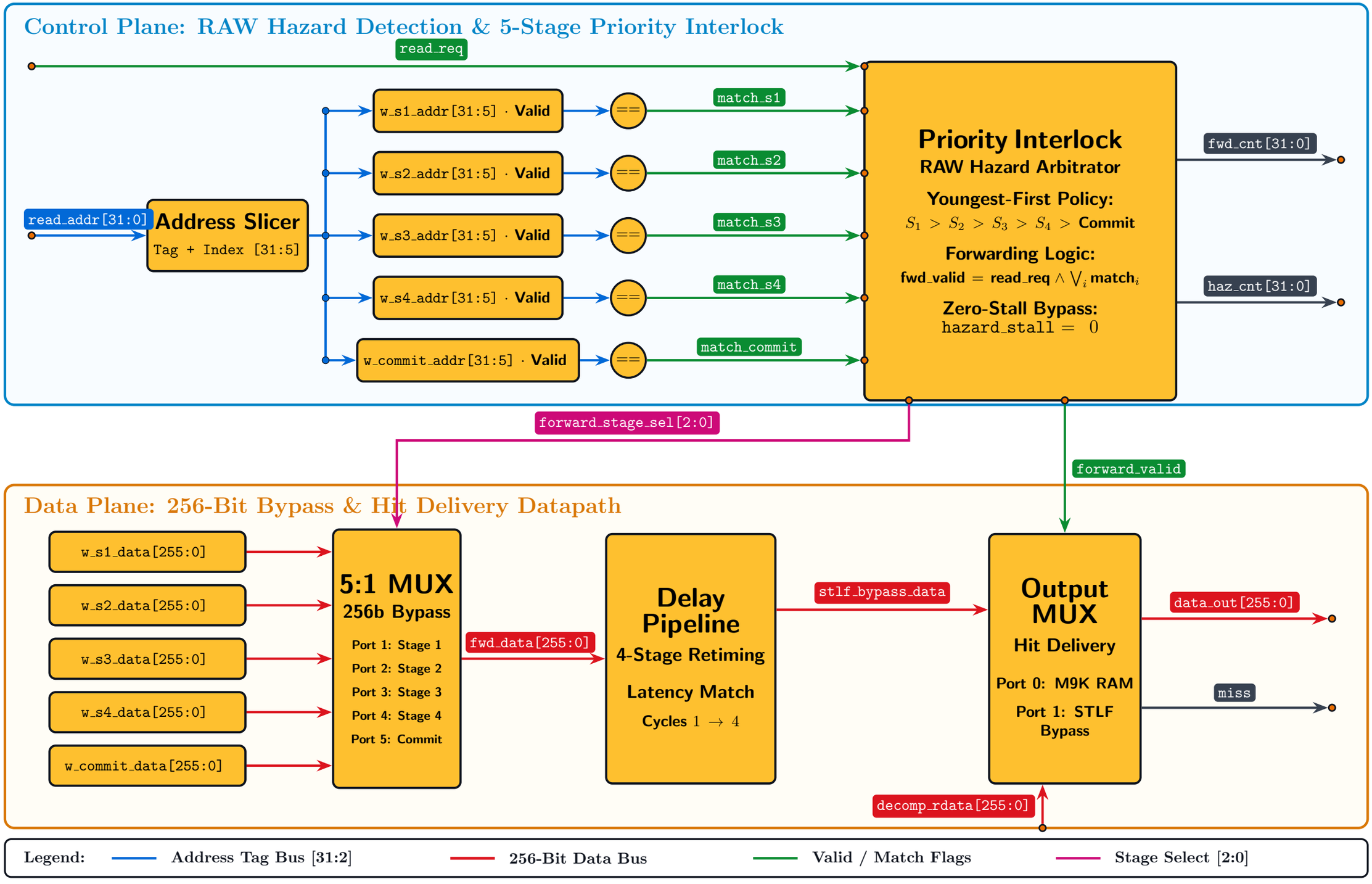

Control Plane: RAW Hazard Detection & 5-Stage Priority Interlock
read_req
read_addr[31:0]
Address Slicer
Tag + Index [31:5]
w_s1_addr[31:5] · Valid
w_s2_addr[31:5] · Valid
w_s3_addr[31:5] · Valid
w_s4_addr[31:5] · Valid
w_commit_addr[31:5] · Valid
==
match_s1
match_s2
match_s3
match_s4
match_commit
Priority Interlock
RAW Hazard Arbitrator
Youngest-First Policy:
$S_1 > S_2 > S_3 > S_4 >$ Commit
Forwarding Logic:
fwd_valid = read_req ∧ ⋁ᵢ matchᵢ
Zero-Stall Bypass:
hazard_stall = 0
fwd_cnt[31:0]
haz_cnt[31:0]
forward_stage_sel[2:0]
forward_valid
Data Plane: 256-Bit Bypass & Hit Delivery Datapath
w_s1_data[255:0]
w_s2_data[255:0]
w_s3_data[255:0]
w_s4_data[255:0]
w_commit_data[255:0]
5:1 MUX
256b Bypass
Port 1: Stage 1
Port 2: Stage 2
Port 3: Stage 3
Port 4: Stage 4
Port 5: Commit
fwd_data[255:0]
Delay Pipeline
4-Stage Retiming
Latency Match
Cycles 1 → 4
stlf_bypass_data
Output MUX
Hit Delivery
Port 0: M9K RAM
Port 1: STLF Bypass
data_out[255:0]
miss
decomp_rdata[255:0]
Legend:
Address Tag Bus [31:2]
256-Bit Data Bus
Valid / Match Flags
Stage Select [2:0]